

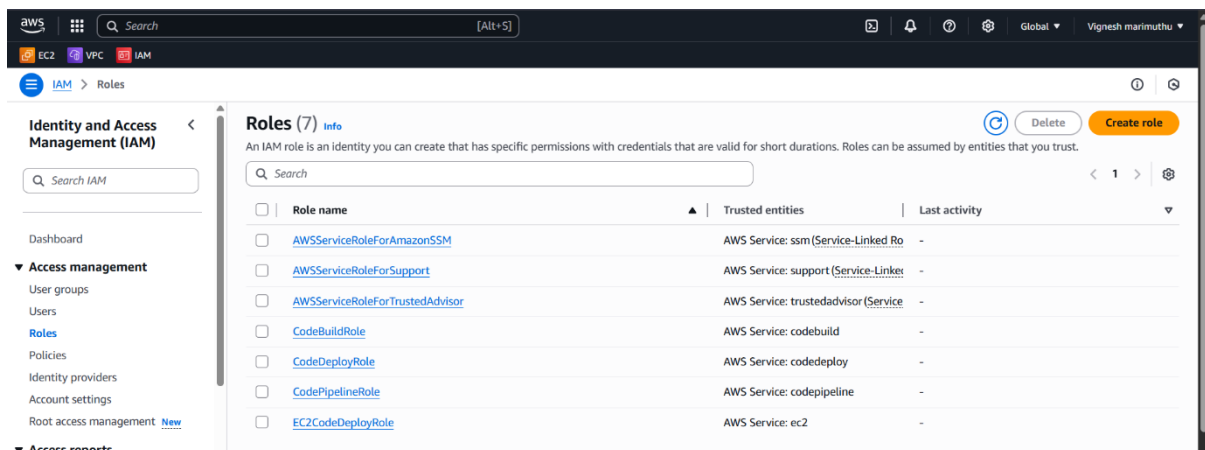
AWS Task-5

Task Description:

Deploy a simple web application using AWS code commit, code build and deploy & access via browser and automate via codepipeline.

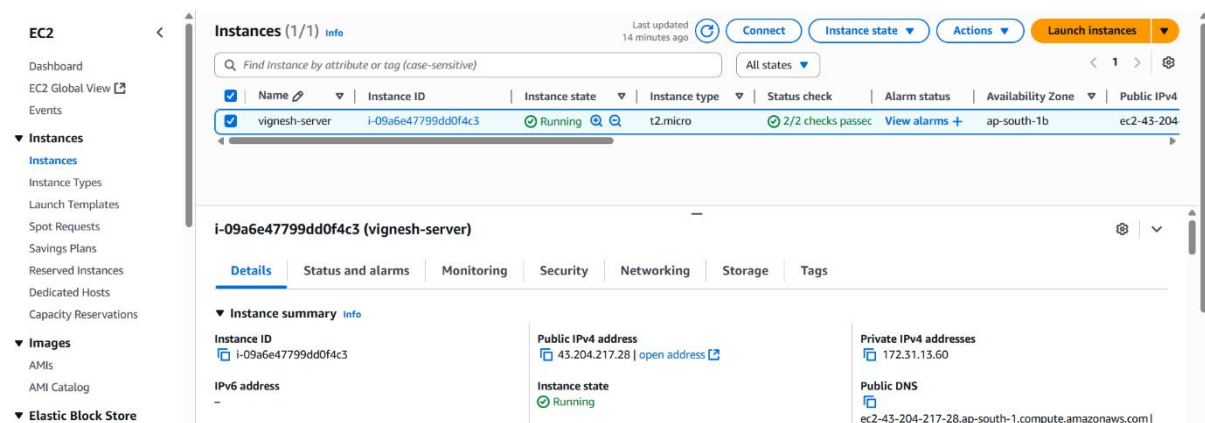
Solution:

✓ 1.Created the IAM roles



- **EC2CodeDeployRole:** Attached AmazonEC2RoleforAWSCodeDeploy
- **CodeDeployRole:** Attached AWSCodeDeployRole
- **CodeBuildRole:** Attached CodeBuildAdminAccess, AmazonS3FullAccess
- **CodePipelineRole:** Custom policy + trust relationship with `codepipeline.amazonaws.com`

✓ 2.launched an EC2 Ubuntu instance



- Opened ports 22, 80 & Attached EC2CodeDeployRole.

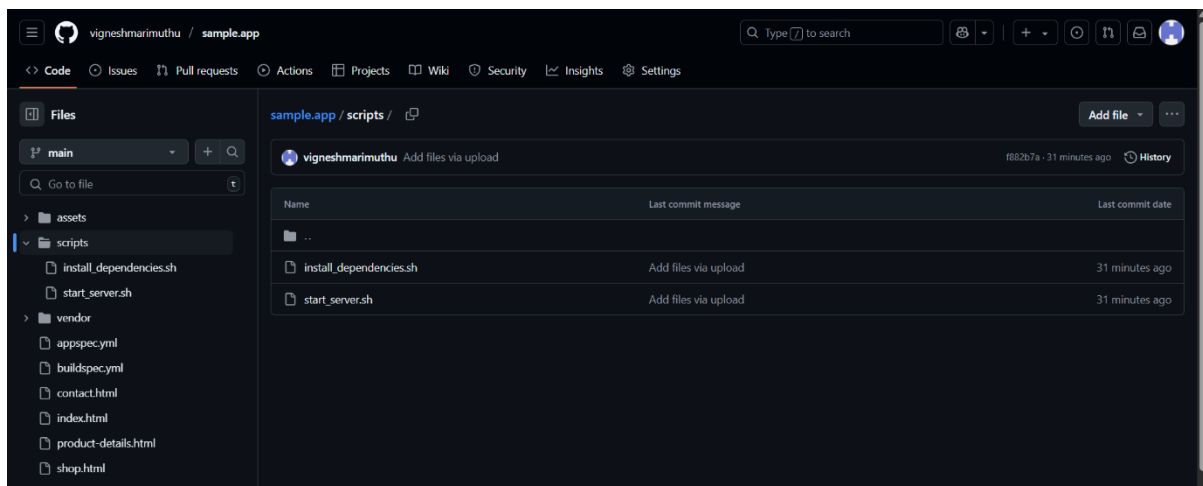
✓ 3. Installed CodeDeploy Agent

```
sudo apt update
sudo apt install ruby-full wget -y
cd /home/ubuntu
wget https://aws-codedeploy-ap-south-1.s3.ap-south-1.amazonaws.com/latest/install
chmod +x install
sudo ./install auto
sudo service codedeploy-agent start

sudo service codedeploy-agent status
```

- SSH into EC2, installed & started the agent using these commands.

✓ 4.Prepared my GitHub repo



- Prepared with index.html, buildspec.yml, appspec.yml, scripts/install_dependencies.sh, scripts/start_server.sh
- My App repo Link : <https://github.com/vigneshmarimuthu/sample.app>

✓ 5.Created buildspec.yml

```
version: 0.2
phases:
  install:
    commands:
      - echo Installing...
  build:
    commands:
      - echo Build phase done...
artifacts:
  files:
    - '**/*'
```

- Contains minimal build config and artifact definition.
- used these commands.

✓ 6.Created appspec.yml

```
version: 0.0
os: linux
files:
  - source: /
    destination: /var/www/html
hooks:
  AfterInstall:
    - location: scripts/install_dependencies.sh
      runas: root
  ApplicationStart:
    - location: scripts/start_server.sh
      runas: root
```

- Mapped source to /var/www/html and defined AfterInstall and ApplicationStart hooks

✓ 7.Created shell scripts

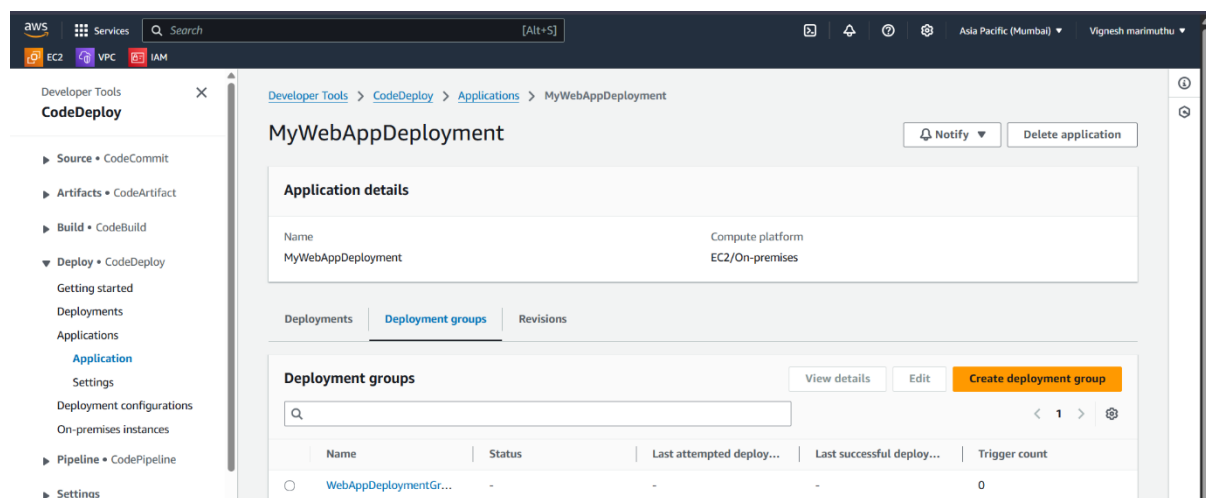
```
#!/bin/bash
apt update -y
apt install apache2 -y
```

- install_dependencies.sh - installs apache2

```
#!/bin/bash
systemctl start apache2
systemctl enable apache2
```

- start_server.sh - starts and enables apache2
- I made the .sh files executable using: **chmod +x scripts/*.sh**

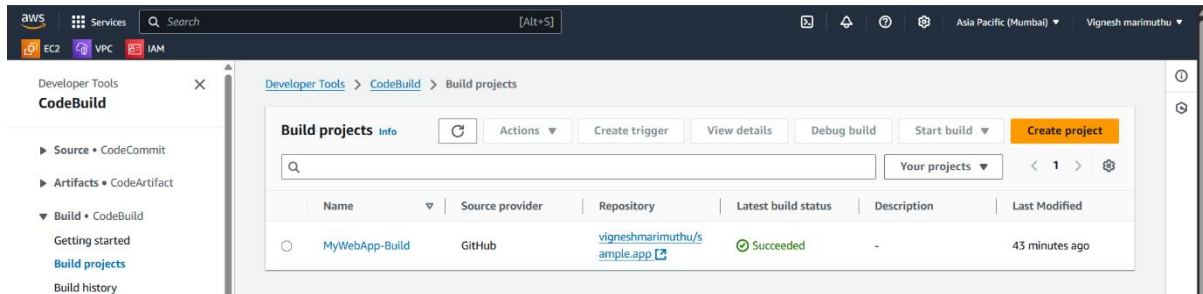
✓ 8. Created CodeDeploy Application & Deployment Group



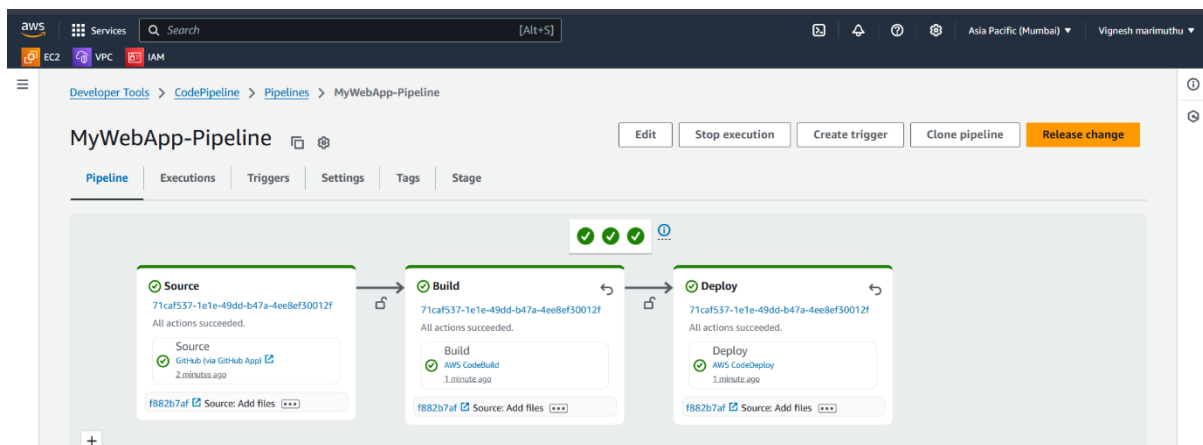
The screenshot shows the AWS CodeDeploy console interface. The left sidebar contains a navigation menu with options like Source, Artifacts, Build, Deploy, Getting started, Deployments, Applications, Settings, Deployment configurations, On-premises instances, Pipeline, and Settings. The main content area displays the 'MyWebAppDeployment' application details, including its name and compute platform (EC2/On-premises). Below this, the 'Deployment groups' tab is active, showing a table with columns for Name, Status, Last attempted deployment, Last successful deployment, and Trigger count. A single deployment group, 'WebAppDeploymentGr...', is listed with a status of '-' and a trigger count of 0. A 'Create deployment group' button is visible in the top right of the deployment groups section.

Name	Status	Last attempted deploy...	Last successful deploy...	Trigger count
WebAppDeploymentGr...	-	-	-	0

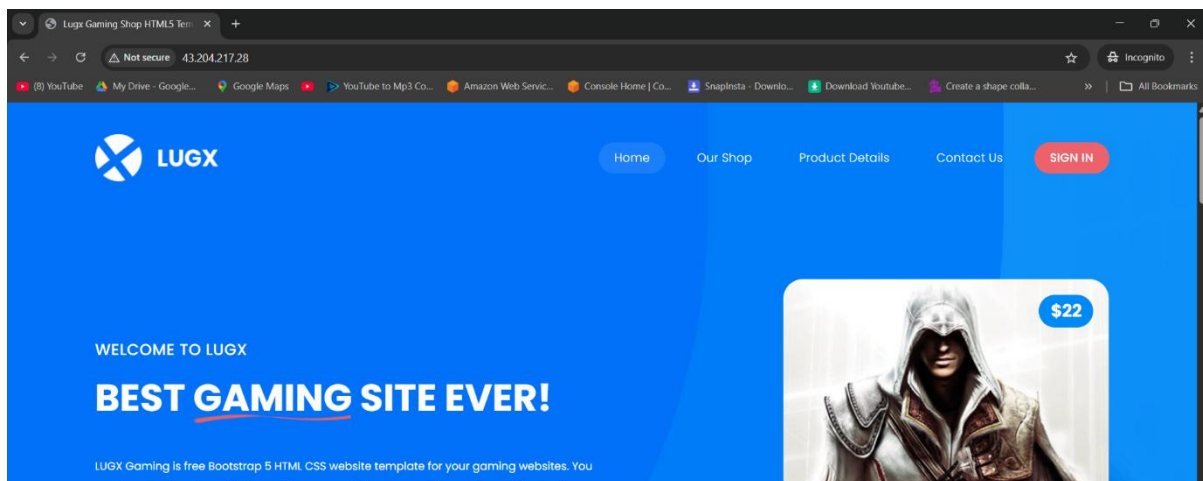
✓ 9.Created a CodeBuild Project



✓ 10.Created a CodePipeline & triggered the pipeline



✓ 11.Verified deployment in browser



- Visited EC2 Public IP and saw the deployed my HTML web page

✓ 12. Verifying Automatic Deployment

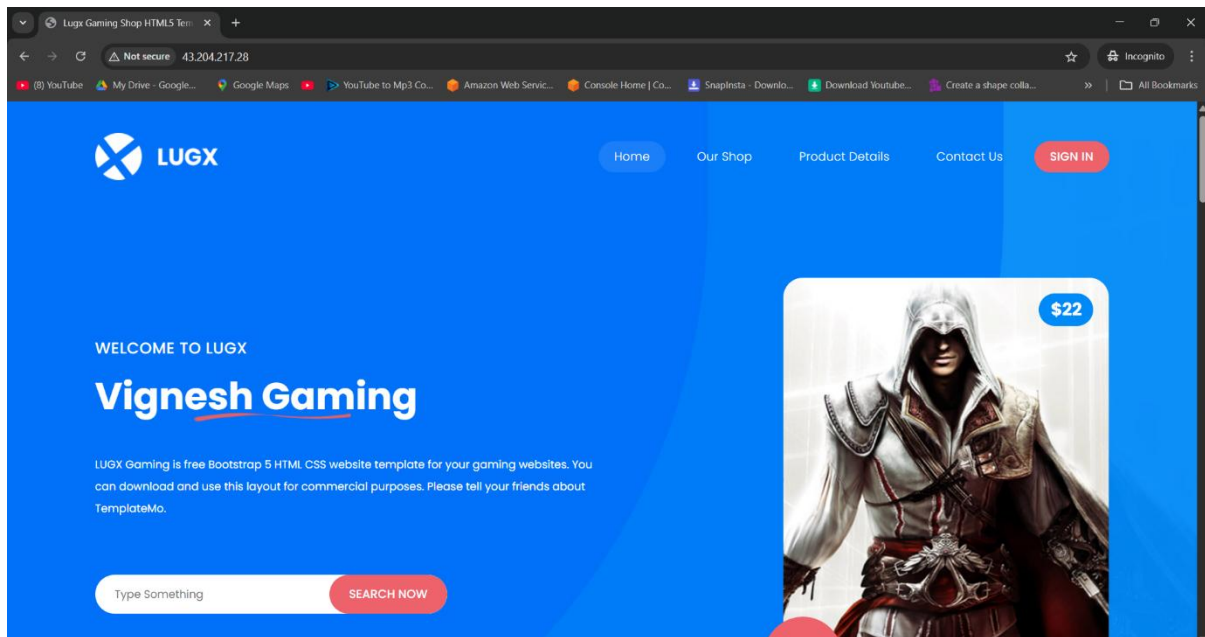
```

72 </div>
73 </header>
74 <!-- ***** Header Area End ***** -->
75
76 <div class="main-banner">
77   <div class="container">
78     <div class="row">
79       <div class="col-lg-6 align-self-center">
80         <div class="caption header-text">
81           <h2>Welcome to lug</h2>
82           <h3>Vignesh Gaming</h3>
83           <p>LUXX Gaming is free Bootstrap 5 HTML CSS website template for your gaming websites. You can download and use this layout for commercial purposes. Ple
84           <div class="search-input">
85             <form id="search" action="#">
86               <input type="text" placeholder="Type Something" id="searchText" name="searchKeyword" onkeypress="handle" />
87               <button role="button">Search Now</button>
88             </form>
89           </div>
90         </div>
91       </div>
92     </div>
93   </div>
94   <div class="col-lg-4 offset-lg-2">

```

Build run	Status	Build number	Source version	Submitter	Duration	Completed
MyWebApp-Build:8e323c57-6711-4b29-b819-c505c063aa5d	Succeeded	9	arn:aws:s3::codepipeline-ap-south-1-f71667859d04-4d60-a337-ff5081978418/MyWebApp-Pipeline/SourceArtifacts/2caf5j6	codepipeline/MyWebApp-Pipeline	8 seconds	Just now
MyWebApp-Build:ade379da-b53c-4519-a32d-de1a83af686c0	Succeeded	8	arn:aws:s3::codepipeline-ap-south-1-f71667859d04-4d60-a337-ff5081978418/MyWebApp-Pipeline/SourceArtifacts/DMnPjSv	codepipeline/MyWebApp-Pipeline	8 seconds	4 minutes ago

Execution ID	Status	Source revisions	Trigger	Started	Duration	Completed
b067e76f	In progress	Source - 58a79b8c: Update index.html	Commit 58a79b8c pushed in vigneshmarimuthu/sample.app/main Update index.html	Jun 23, 2025 7:07 AM (UTC+5:30)	0 seconds	-
71caf537	Succeeded	Source - f882b7af: Add files via upload	StartPipelineExecution - root	Jun 23, 2025 7:03 AM (UTC+5:30)	1 minute 21 seconds	Jun 23, 2025 7:04 AM (UTC+5:30)
7d5cb2ab	Succeeded	Source - f882b7af: Add files via upload	Commit f882b7af pushed in vigneshmarimuthu/sample.app/main Add files via upload	Jun 23, 2025 7:02 AM (UTC+5:30)	1 minute 9 seconds	Jun 23, 2025 7:04 AM (UTC+5:30)



- After I successfully deployed the application, I made changes to the HTML code in my GitHub repo. This automatically triggered the pipeline, which rebuilt and redeployed the updated code to my EC2 instance
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